

TABLE 01

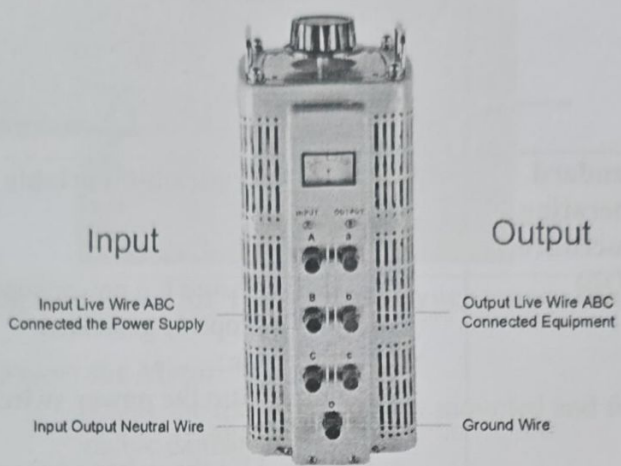
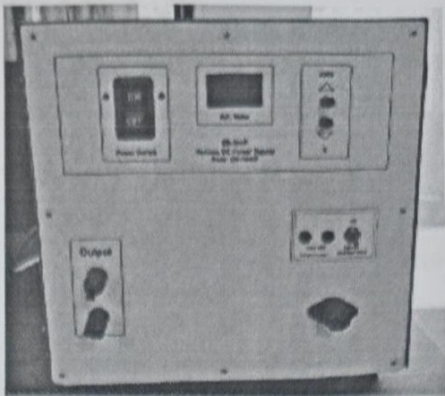
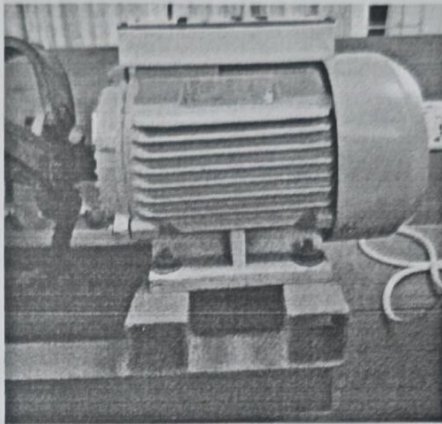
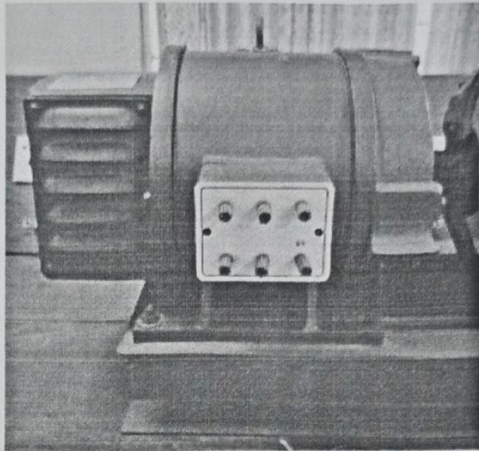
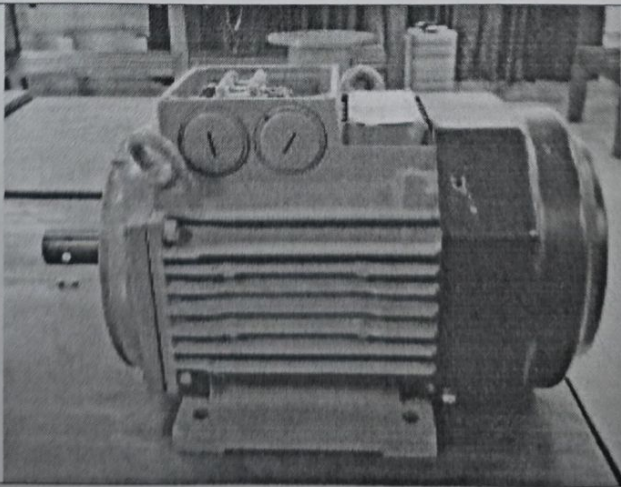
01.	Component name	Three-phase VARIAC
	Quantity	01
	Ratings	Input: 430-440V, 50HZ Output: 0-430V, 50HZ
	Connection Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase VARIAC: 1. Inspect the VARIAC for damages and ensure it is grounded. 2. Connect Input Wires: Attach live wires (A, B, C) and neutral wire (O) to the input terminals. 3. Connect Output Wires: Attach live wires (A, B, C) and ground wire to the output terminals. 4. Adjust Voltage: Rotate the control knob to set the desired output voltage. 5. Power On: Switch on the input power supply and monitor voltage using the meter. 6. Operate Equipment: Connect and operate the equipment after stabilizing the output voltage. 7. Shut Down: Turn off equipment, power off the VARIAC, and disconnect wires. 8. Maintain: Regularly clean, check for loose connections, and inspect for wear.
02.	Component name	Variable DC Power Supply
	Quantity	01
	Ratings	Input: 220V Output Variable: 0-220V

	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Variable DC Power Supply: Inspect the Device: - Ensure the power supply is in good condition and properly grounded. Power On: - Turn the power switch to the "ON" position. Set Voltage: - Use the voltage adjustment knob or buttons to set the desired DC voltage. - Confirm the voltage level on the digital voltmeter display. Connect Load: - Attach the positive and negative terminals of the output to the respective terminals of the load or equipment. Operate Equipment: - Start the connected equipment and monitor the voltage and current levels during operation. Adjust as Needed: - Fine-tune the voltage or current using the controls, if required, during operation. Shut Down: - Turn off the connected equipment. - Switch the power supply off using the power switch. - Disconnect the output terminals. Maintenance: - Periodically clean the device and check for loose connections or wear.
03.	Component name	Three-phase synchronous motor
	Quantity	01
	Ratings	P=2200W

	PF: 01 F=50Hz n=1500 RPM
Diagram	
Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Synchronous Motor: Inspect the Motor: - Ensure the motor is securely mounted and free from visible damage. - Check that all connections are properly secured. Connect Power Supply: - Attach the three-phase power supply wires to the respective terminals of the motor. - Ensure proper grounding of the motor. Check Auxiliary Equipment: - Confirm the exciter (if applicable) is operational and properly connected. Start the Motor: - Apply the three-phase power supply. - Use a starter or control panel to gradually bring the motor to synchronous speed. Synchronize the Motor: - Adjust the excitation to achieve the desired power factor and synchronize with the power supply. Monitor Operation: - Continuously monitor speed, voltage, and current to ensure stable operation. - Listen for unusual sounds or vibrations during operation. Shut Down: - Turn off the power supply and excitation system. - Allow the motor to come to a complete stop before disconnecting any connections. Maintenance: - Regularly inspect and clean the motor.

		○ Lubricate bearings as per the manufacturer's guidelines. ○ Check winding insulation and connections periodically.
04.	Component name	Three-phase synchronous generator
	Quantity	01
	Ratings	P=1800VA PF: 01 f=50Hz n=1500 RPM
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Synchronous Generator: 1. Inspect the Generator: ○ Ensure the generator is securely mounted and free from physical damage. ○ Check that all terminals and connections are intact. 2. Connect Load: ○ Attach the load wires to the respective terminals of the generator. ○ Ensure the ground wire is properly connected. 3. Connect Prime Mover: ○ Ensure the generator is mechanically coupled to a prime mover (e.g., turbine or engine). ○ Check for proper alignment and secure coupling. 4. Start the Prime Mover: ○ Gradually increase the speed of the prime mover to bring the generator to its rated speed. 5. Excitation: ○ Apply the required field excitation to establish the magnetic field.

		○ Adjust excitation to regulate voltage output. 6. Synchronize with the Grid (if applicable): ○ Match the generator's voltage, frequency, and phase sequence with the grid before connection. ○ Close the circuit breaker to connect the generator to the grid. 7. Monitor Operation: ○ Continuously monitor voltage, current, and frequency to ensure stable operation. ○ Check for unusual sounds or vibrations. 8. Shut Down: ○ Disconnect the generator from the grid or load. ○ Gradually reduce the prime mover speed to bring the generator to a complete stop. ○ Turn off the field excitation. 9. Maintenance: ○ Clean and inspect the generator regularly. ○ Check windings, connections, and insulation for wear or damage. ○ Lubricate bearings as recommended by the manufacturer.
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05.	Component name	Three-phase induction motor
	Quantity	01
	Ratings	Voltage: 415 ± 10 V $f = 50 \pm 5\%$ Hz $n = 1430$ RPM P: 2.2KW Horse power: 3 H.P. Current: 4.6A P.F.: 0.83 Efficiency: 82.80%
	Diagram	

	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Induction Motor: Inspect the Motor: - Ensure the motor is securely mounted and free from visible damage. - Check that all terminals and connections are in proper condition. Connect Power Supply: - Attach the three-phase power supply wires to the motor terminals. - Ensure the motor is properly grounded. Check Starter: - Connect a suitable starter (e.g., DOL starter, star-delta starter) for controlled motor starting. Start the Motor: - Switch on the power supply through the starter. - Gradually bring the motor up to its rated speed. Monitor Operation: - Continuously observe voltage, current, and motor speed. - Check for unusual noises, vibrations, or overheating during operation. Adjust Load: - Ensure the load connected to the motor is within its rated capacity. - Adjust the load as necessary for optimal performance. Shut Down: - Turn off the power supply via the starter. - Allow the motor to come to a complete stop before disconnecting the load or supply. Maintenance: - Regularly clean the motor and remove dust or debris. - Inspect bearings and lubricate as needed. - Check winding insulation and terminal connections periodically.
06.	Component name	Single-phase Transformer
	Quantity	01
	Ratings	Rated power 10 - 2500VA Rated input voltage < 1000V Rated output voltage < 500V Frequency 50/60Hz

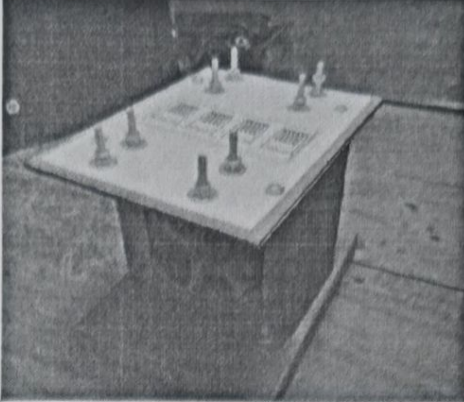
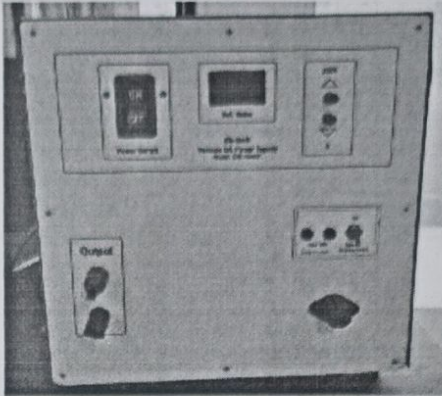
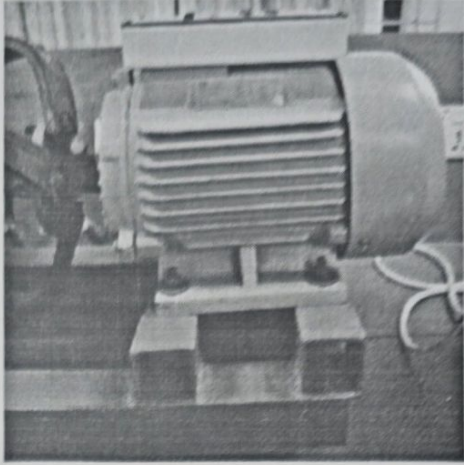
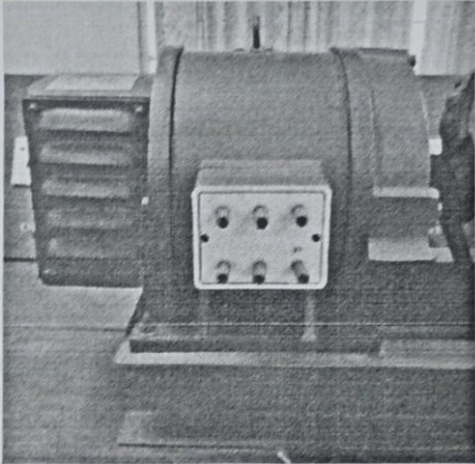
Diagram	
Standard Operating Procedure (SOP)	Operating Procedure for Single-phase Transformer: 1. Inspect the Transformer: ○ Check for physical damage and ensure the transformer is properly mounted. ○ Verify that all terminals are clean and secure. 2. Connect the Input (Primary Side): ○ Attach the primary winding terminals to the appropriate AC power source. ○ Ensure the voltage matches the transformer's input rating. 3. Connect the Output (Secondary Side): ○ Connect the load to the secondary winding terminals as per the required voltage. ○ Confirm that the load does not exceed the rated capacity of the transformer. 4. Grounding: ○ Ensure the transformer is properly grounded for safety. 5. Power On: ○ Switch on the power supply to the primary winding. ○ Observe the output voltage to ensure it matches the expected level. 6. Monitor Operation: ○ Continuously monitor the transformer's temperature and output voltage. ○ Check for unusual sounds, vibrations, or overheating. 7. Shut Down: ○ Turn off the power supply to the primary winding. ○ Disconnect the load and input connections if needed. 8. Maintenance: ○ Periodically clean the transformer to remove dust and debris. ○ Inspect the windings, terminals, and insulation for wear or damage.

TABLE 02	
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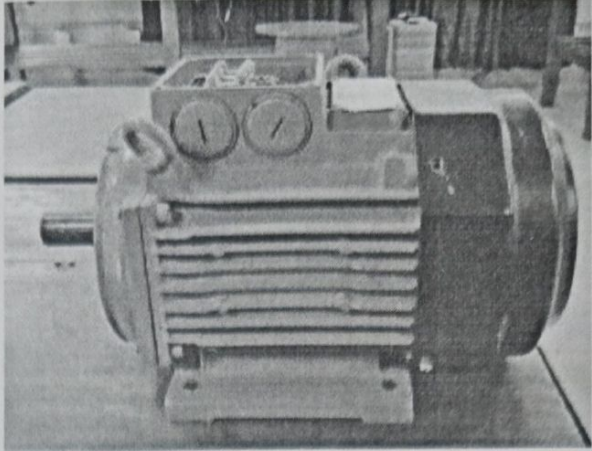
01.	Component name	Three-phase VARIAC
	Quantity	01
	Ratings	Input: 430-440V, 50HZ Output: 0-430V, 50HZ
	Connection Diagram	Input Input Live Wire ABC Connected the Power Supply Output Output Live Wire ABC Connected Equipment Input Output Neutral Wire Ground Wire
02.	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase VARIAC: 1. Inspect the VARIAC for damages and ensure it is grounded. 2. Connect Input Wires: Attach live wires (A, B, C) and neutral wire (O) to the input terminals. 3. Connect Output Wires: Attach live wires (A, B, C) and ground wire to the output terminals. 4. Adjust Voltage: Rotate the control knob to set the desired output voltage. 5. Power On: Switch on the input power supply and monitor voltage using the meter. 6. Operate Equipment: Connect and operate the equipment after stabilizing the output voltage. 7. Shut Down: Turn off equipment, power off the VARIAC, and disconnect wires. 8. Maintain: Regularly clean, check for loose connections, and inspect for wear.
	Component name	Variable DC Power Supply
	Quantity	01

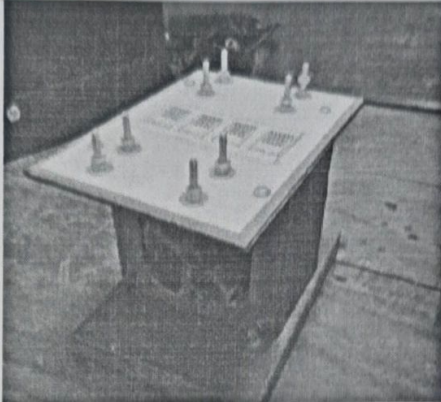
	Ratings	Input: 220V Output Variable: 0-220V
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Variable DC Power Supply: Inspect the Device: - Ensure the power supply is in good condition and properly grounded. Power On: - Turn the power switch to the "ON" position. Set Voltage: - Use the voltage adjustment knob or buttons to set the desired DC voltage. - Confirm the voltage level on the digital voltmeter display. Connect Load: - Attach the positive and negative terminals of the output to the respective terminals of the load or equipment. Operate Equipment: - Start the connected equipment and monitor the voltage and current levels during operation. Adjust as Needed: - Fine-tune the voltage or current using the controls, if required, during operation. Shut Down: - Turn off the connected equipment. - Switch the power supply off using the power switch. - Disconnect the output terminals. Maintenance: - Periodically clean the device and check for loose connections or wear.
03.	Component name	Three-phase synchronous motor

Quantity	01
Ratings	P=2200W PF: 01 F=50Hz n=1500 RPM
Diagram	
Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Synchronous Motor: 1. Inspect the Motor: ○ Ensure the motor is securely mounted and free from visible damage. ○ Check that all connections are properly secured. 2. Connect Power Supply: ○ Attach the three-phase power supply wires to the respective terminals of the motor. ○ Ensure proper grounding of the motor. 3. Check Auxiliary Equipment: ○ Confirm the exciter (if applicable) is operational and properly connected. 4. Start the Motor: ○ Apply the three-phase power supply. ○ Use a starter or control panel to gradually bring the motor to synchronous speed. 5. Synchronize the Motor: ○ Adjust the excitation to achieve the desired power factor and synchronize with the power supply. 6. Monitor Operation: ○ Continuously monitor speed, voltage, and current to ensure stable operation. ○ Listen for unusual sounds or vibrations during operation. 7. Shut Down: ○ Turn off the power supply and excitation system.

		○ Allow the motor to come to a complete stop before disconnecting any connections. 8. Maintenance: ○ Regularly inspect and clean the motor. ○ Lubricate bearings as per the manufacturer's guidelines. ○ Check winding insulation and connections periodically.
04.	Component name	Three-phase synchronous generator
	Quantity	01
	Ratings	P=1800VA PF: 01 f=50Hz n=1500 RPM
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Synchronous Generator: 1. Inspect the Generator: ○ Ensure the generator is securely mounted and free from physical damage. ○ Check that all terminals and connections are intact. 2. Connect Load: ○ Attach the load wires to the respective terminals of the generator. ○ Ensure the ground wire is properly connected. 3. Connect Prime Mover: ○ Ensure the generator is mechanically coupled to a prime mover (e.g., turbine or engine). ○ Check for proper alignment and secure coupling. 4. Start the Prime Mover:

		○ Gradually increase the speed of the prime mover to bring the generator to its rated speed. 5. Excitation: ○ Apply the required field excitation to establish the magnetic field. ○ Adjust excitation to regulate voltage output. 6. Synchronize with the Grid (if applicable): ○ Match the generator's voltage, frequency, and phase sequence with the grid before connection. ○ Close the circuit breaker to connect the generator to the grid. 7. Monitor Operation: ○ Continuously monitor voltage, current, and frequency to ensure stable operation. ○ Check for unusual sounds or vibrations. 8. Shut Down: ○ Disconnect the generator from the grid or load. ○ Gradually reduce the prime mover speed to bring the generator to a complete stop. ○ Turn off the field excitation. 9. Maintenance: ○ Clean and inspect the generator regularly. ○ Check windings, connections, and insulation for wear or damage. ○ Lubricate bearings as recommended by the manufacturer.
05.	Component name	Three-phase induction motor
	Quantity	01
	Ratings	Voltage: 415±10 V f=50±5% Hz n=1430 RPM P: 2.2KW Horse power: 3 H.P. Current: 4.6A P.F.: 0.83 Efficiency: 82.80%

Diagram	
Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Induction Motor: 1. Inspect the Motor: ○ Ensure the motor is securely mounted and free from visible damage. ○ Check that all terminals and connections are in proper condition. 2. Connect Power Supply: ○ Attach the three-phase power supply wires to the motor terminals. ○ Ensure the motor is properly grounded. 3. Check Starter: ○ Connect a suitable starter (e.g., DOL starter, star-delta starter) for controlled motor starting. 4. Start the Motor: ○ Switch on the power supply through the starter. ○ Gradually bring the motor up to its rated speed. 5. Monitor Operation: ○ Continuously observe voltage, current, and motor speed. ○ Check for unusual noises, vibrations, or overheating during operation. 6. Adjust Load: ○ Ensure the load connected to the motor is within its rated capacity. ○ Adjust the load as necessary for optimal performance. 7. Shut Down: ○ Turn off the power supply via the starter. ○ Allow the motor to come to a complete stop before disconnecting the load or supply. 8. Maintenance: ○ Regularly clean the motor and remove dust or debris. ○ Inspect bearings and lubricate as needed.

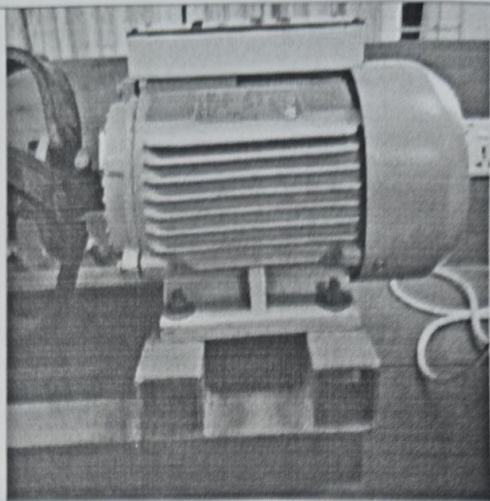
		○ Check winding insulation and terminal connections periodically.
06.	Component name	Single-phase Transformer
	Quantity	01
	Ratings	Rated power 10 - 2500VA Rated input voltage < 1000V Rated output voltage < 500V Frequency 50/60Hz
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Single-phase Transformer: 1. Inspect the Transformer: ○ Check for physical damage and ensure the transformer is properly mounted. ○ Verify that all terminals are clean and secure. 2. Connect the Input (Primary Side): ○ Attach the primary winding terminals to the appropriate AC power source. ○ Ensure the voltage matches the transformer's input rating. 3. Connect the Output (Secondary Side): ○ Connect the load to the secondary winding terminals as per the required voltage. ○ Confirm that the load does not exceed the rated capacity of the transformer. 4. Grounding: ○ Ensure the transformer is properly grounded for safety. 5. Power On: ○ Switch on the power supply to the primary winding. ○ Observe the output voltage to ensure it matches the expected level. 6. Monitor Operation: ○ Continuously monitor the transformer's temperature and output voltage.

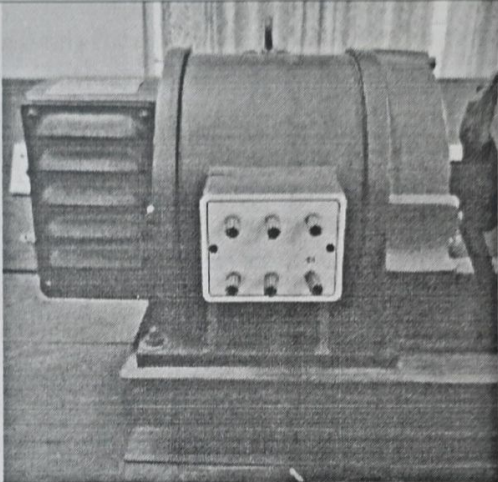
		○ Check for unusual sounds, vibrations, or overheating. 7. Shut Down: ○ Turn off the power supply to the primary winding. ○ Disconnect the load and input connections if needed. 8. Maintenance: ○ Periodically clean the transformer to remove dust and debris. ○ Inspect the windings, terminals, and insulation for wear or damage.
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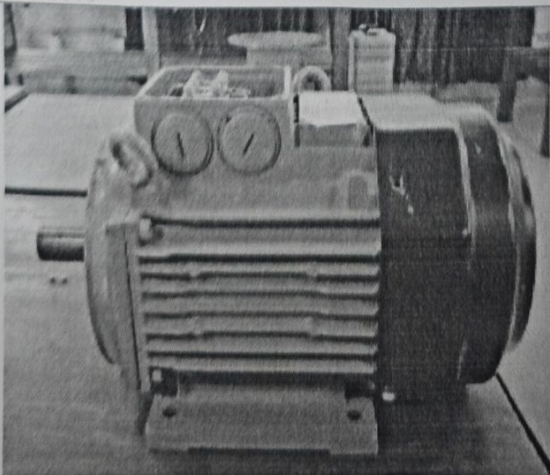
TABLE 03

01.	Component name	Three-phase VARIAC
	Quantity	01
	Ratings	Input: 430-440V, 50HZ Output: 0-430V, 50HZ
	Connection Diagram	Input Output Input Live Wire ABC Connected the Power Supply Output Live Wire ABC Connected Equipment Input Output Neutral Wire Ground Wire
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase VARIAC: 1. Inspect the VARIAC for damages and ensure it is grounded. 2. Connect Input Wires: Attach live wires (A, B, C) and neutral wire (O) to the input terminals. 3. Connect Output Wires: Attach live wires (A, B, C) and ground wire to the output terminals. 4. Adjust Voltage: Rotate the control knob to set the desired output voltage. 5. Power On: Switch on the input power supply and monitor voltage using the meter. 6. Operate Equipment: Connect and operate the equipment after stabilizing the output voltage. 7. Shut Down: Turn off equipment, power off the VARIAC, and disconnect wires. 8. Maintain: Regularly clean, check for loose connections, and inspect for wear.
02.	Component name	Variable DC Power Supply
	Quantity	01
	Ratings	Input: 220V Output Variable: 0-220V

Diagram							
Standard Operating Procedure (SOP)	Operating Procedure for Variable DC Power Supply: Inspect the Device: - Ensure the power supply is in good condition and properly grounded. Power On: - Turn the power switch to the "ON" position. Set Voltage: - Use the voltage adjustment knob or buttons to set the desired DC voltage. - Confirm the voltage level on the digital voltmeter display. Connect Load: - Attach the positive and negative terminals of the output to the respective terminals of the load or equipment. Operate Equipment: - Start the connected equipment and monitor the voltage and current levels during operation. Adjust as Needed: - Fine-tune the voltage or current using the controls, if required, during operation. Shut Down: - Turn off the connected equipment. - Switch the power supply off using the power switch. - Disconnect the output terminals. Maintenance: - Periodically clean the device and check for loose connections or wear.						
03.	<table border="1"> <tr> <td data-bbox="365 1690 803 1766">Component name</td><td data-bbox="803 1690 1307 1766">Three-phase synchronous motor</td></tr> <tr> <td data-bbox="365 1766 803 1816">Quantity</td><td data-bbox="803 1766 1307 1816">01</td></tr> <tr> <td data-bbox="365 1816 803 1885">Ratings</td><td data-bbox="803 1816 1307 1885">P=2200W PF: 01</td></tr> </table>	Component name	Three-phase synchronous motor	Quantity	01	Ratings	P=2200W PF: 01
Component name	Three-phase synchronous motor						
Quantity	01						
Ratings	P=2200W PF: 01						

	F=50Hz n=1500 RPM	
Diagram		
Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Synchronous Motor: Inspect the Motor: - Ensure the motor is securely mounted and free from visible damage. - Check that all connections are properly secured. Connect Power Supply: - Attach the three-phase power supply wires to the respective terminals of the motor. - Ensure proper grounding of the motor. Check Auxiliary Equipment: - Confirm the exciter (if applicable) is operational and properly connected. Start the Motor: - Apply the three-phase power supply. - Use a starter or control panel to gradually bring the motor to synchronous speed. Synchronize the Motor: - Adjust the excitation to achieve the desired power factor and synchronize with the power supply. Monitor Operation: - Continuously monitor speed, voltage, and current to ensure stable operation. - Listen for unusual sounds or vibrations during operation. Shut Down: - Turn off the power supply and excitation system. - Allow the motor to come to a complete stop before disconnecting any connections. Maintenance: - Regularly inspect and clean the motor.	

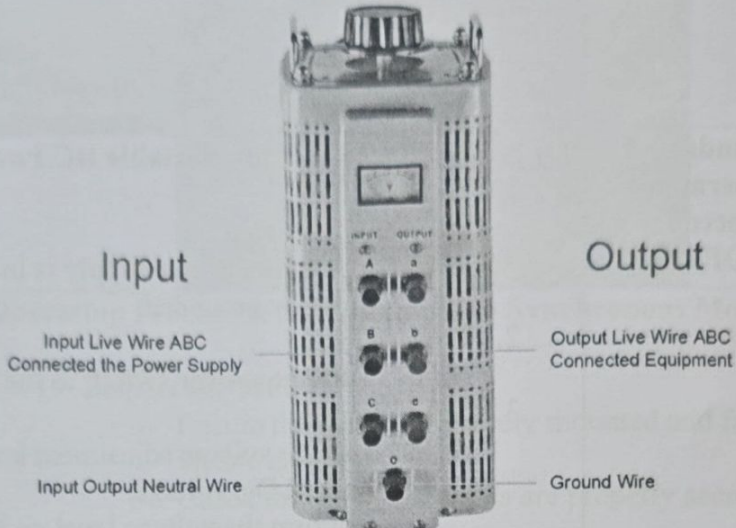
		○ Lubricate bearings as per the manufacturer's guidelines. ○ Check winding insulation and connections periodically.
04.	Component name	Three-phase synchronous generator
	Quantity	01
	Ratings	P=1800VA PF: 01 f=50Hz n=1500 RPM
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Synchronous Generator: 1. Inspect the Generator: ○ Ensure the generator is securely mounted and free from physical damage. ○ Check that all terminals and connections are intact. 2. Connect Load: ○ Attach the load wires to the respective terminals of the generator. ○ Ensure the ground wire is properly connected. 3. Connect Prime Mover: ○ Ensure the generator is mechanically coupled to a prime mover (e.g., turbine or engine). ○ Check for proper alignment and secure coupling. 4. Start the Prime Mover: ○ Gradually increase the speed of the prime mover to bring the generator to its rated speed. 5. Excitation: ○ Apply the required field excitation to establish the magnetic field. ○ Adjust excitation to regulate voltage output.

		6. Synchronize with the Grid (if applicable): ○ Match the generator's voltage, frequency, and phase sequence with the grid before connection. ○ Close the circuit breaker to connect the generator to the grid. 7. Monitor Operation: ○ Continuously monitor voltage, current, and frequency to ensure stable operation. ○ Check for unusual sounds or vibrations. 8. Shut Down: ○ Disconnect the generator from the grid or load. ○ Gradually reduce the prime mover speed to bring the generator to a complete stop. ○ Turn off the field excitation. 9. Maintenance: ○ Clean and inspect the generator regularly. ○ Check windings, connections, and insulation for wear or damage. ○ Lubricate bearings as recommended by the manufacturer.
05.	Component name	Three-phase induction motor
	Quantity	01
	Ratings	Voltage: 415 ± 10 V $f = 50 \pm 5\%$ Hz $n = 1430$ RPM P: 2.2KW Horse power: 3 H.P. Current: 4.6A P.F.: 0.83 Efficiency: 82.80%
	Diagram	

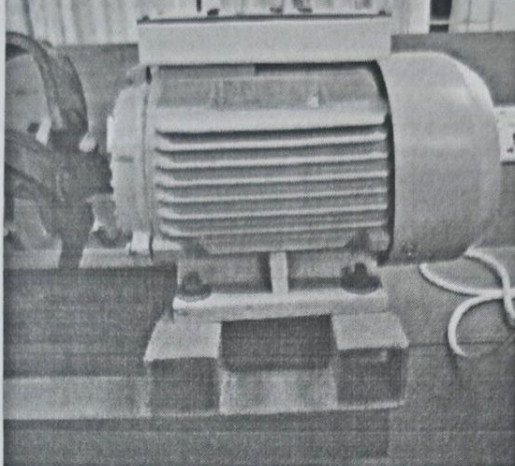
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Induction Motor: 1. Inspect the Motor: ○ Ensure the motor is securely mounted and free from visible damage. ○ Check that all terminals and connections are in proper condition. 2. Connect Power Supply: ○ Attach the three-phase power supply wires to the motor terminals. ○ Ensure the motor is properly grounded. 3. Check Starter: ○ Connect a suitable starter (e.g., DOL starter, star-delta starter) for controlled motor starting. 4. Start the Motor: ○ Switch on the power supply through the starter. ○ Gradually bring the motor up to its rated speed. 5. Monitor Operation: ○ Continuously observe voltage, current, and motor speed. ○ Check for unusual noises, vibrations, or overheating during operation. 6. Adjust Load: ○ Ensure the load connected to the motor is within its rated capacity. ○ Adjust the load as necessary for optimal performance. 7. Shut Down: ○ Turn off the power supply via the starter. ○ Allow the motor to come to a complete stop before disconnecting the load or supply. 8. Maintenance: ○ Regularly clean the motor and remove dust or debris. ○ Inspect bearings and lubricate as needed. ○ Check winding insulation and terminal connections periodically.
06.	Component name	Single-phase Transformer
	Quantity	01
	Ratings	Rated power 10 - 2500VA Rated input voltage < 1000V Rated output voltage < 500V Frequency 50/60Hz

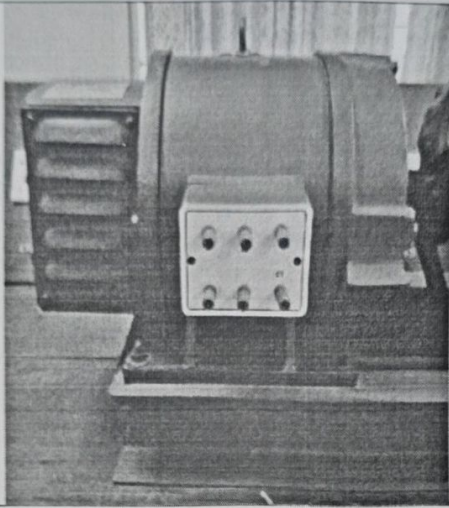
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Single-phase Transformer: 1. Inspect the Transformer: ○ Check for physical damage and ensure the transformer is properly mounted. ○ Verify that all terminals are clean and secure. 2. Connect the Input (Primary Side): ○ Attach the primary winding terminals to the appropriate AC power source. ○ Ensure the voltage matches the transformer's input rating. 3. Connect the Output (Secondary Side): ○ Connect the load to the secondary winding terminals as per the required voltage. ○ Confirm that the load does not exceed the rated capacity of the transformer. 4. Grounding: ○ Ensure the transformer is properly grounded for safety. 5. Power On: ○ Switch on the power supply to the primary winding. ○ Observe the output voltage to ensure it matches the expected level. 6. Monitor Operation: ○ Continuously monitor the transformer's temperature and output voltage. ○ Check for unusual sounds, vibrations, or overheating. 7. Shut Down: ○ Turn off the power supply to the primary winding. ○ Disconnect the load and input connections if needed. 8. Maintenance: ○ Periodically clean the transformer to remove dust and debris. ○ Inspect the windings, terminals, and insulation for wear or damage.

TABLE 04

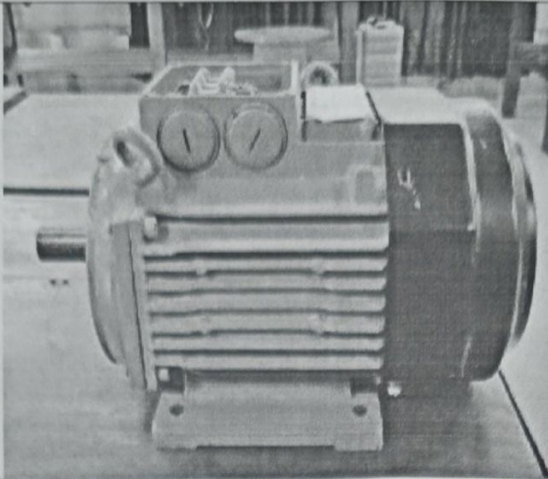
01.	Component name	Three-phase VARIAC
	Quantity	01
	Ratings	Input: 430-440V, 50HZ Output: 0-430V, 50HZ
	Connection Diagram	 Input Output Input Live Wire ABC Connected the Power Supply Output Live Wire ABC Connected Equipment Input Output Neutral Wire Ground Wire
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase VARIAC: 1. Inspect the VARIAC for damages and ensure it is grounded. 2. Connect Input Wires: Attach live wires (A, B, C) and neutral wire (O) to the input terminals. 3. Connect Output Wires: Attach live wires (A, B, C) and ground wire to the output terminals. 4. Adjust Voltage: Rotate the control knob to set the desired output voltage. 5. Power On: Switch on the input power supply and monitor voltage using the meter. 6. Operate Equipment: Connect and operate the equipment after stabilizing the output voltage. 7. Shut Down: Turn off equipment, power off the VARIAC, and disconnect wires. 8. Maintain: Regularly clean, check for loose connections, and inspect for wear.
02.	Component name	Variable DC Power Supply
	Quantity	01

	Ratings	Input: 220V Output Variable: 0-220V
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Variable DC Power Supply: Inspect the Device: - Ensure the power supply is in good condition and properly grounded. Power On: - Turn the power switch to the "ON" position. Set Voltage: - Use the voltage adjustment knob or buttons to set the desired DC voltage. - Confirm the voltage level on the digital voltmeter display. Connect Load: - Attach the positive and negative terminals of the output to the respective terminals of the load or equipment. Operate Equipment: - Start the connected equipment and monitor the voltage and current levels during operation. Adjust as Needed: - Fine-tune the voltage or current using the controls, if required, during operation. Shut Down: - Turn off the connected equipment. - Switch the power supply off using the power switch. - Disconnect the output terminals. Maintenance: - Periodically clean the device and check for loose connections or wear.
03.	Component name	Three-phase synchronous motor
	Quantity	01

Ratings	P=2200W PF: 01 F=50Hz n=1500 RPM
Diagram	
Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Synchronous Motor: Inspect the Motor: - Ensure the motor is securely mounted and free from visible damage. - Check that all connections are properly secured. Connect Power Supply: - Attach the three-phase power supply wires to the respective terminals of the motor. - Ensure proper grounding of the motor. Check Auxiliary Equipment: - Confirm the exciter (if applicable) is operational and properly connected. Start the Motor: - Apply the three-phase power supply. - Use a starter or control panel to gradually bring the motor to synchronous speed. Synchronize the Motor: - Adjust the excitation to achieve the desired power factor and synchronize with the power supply. Monitor Operation: - Continuously monitor speed, voltage, and current to ensure stable operation. - Listen for unusual sounds or vibrations during operation. Shut Down: - Turn off the power supply and excitation system. - Allow the motor to come to a complete stop before disconnecting any connections.

		8. Maintenance: ○ Regularly inspect and clean the motor. ○ Lubricate bearings as per the manufacturer's guidelines. ○ Check winding insulation and connections periodically.
04.	Component name	Three-phase synchronous generator
	Quantity	01
	Ratings	P=1800VA PF: 01 f=50Hz n=1500 RPM
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Synchronous Generator: Inspect the Generator: ○ Ensure the generator is securely mounted and free from physical damage. ○ Check that all terminals and connections are intact. Connect Load: ○ Attach the load wires to the respective terminals of the generator. ○ Ensure the ground wire is properly connected. Connect Prime Mover: ○ Ensure the generator is mechanically coupled to a prime mover (e.g., turbine or engine). ○ Check for proper alignment and secure coupling. Start the Prime Mover: ○ Gradually increase the speed of the prime mover to bring the generator to its rated speed. Excitation:

		○ Apply the required field excitation to establish the magnetic field. ○ Adjust excitation to regulate voltage output. 6. Synchronize with the Grid (if applicable): ○ Match the generator's voltage, frequency, and phase sequence with the grid before connection. ○ Close the circuit breaker to connect the generator to the grid. 7. Monitor Operation: ○ Continuously monitor voltage, current, and frequency to ensure stable operation. ○ Check for unusual sounds or vibrations. 8. Shut Down: ○ Disconnect the generator from the grid or load. ○ Gradually reduce the prime mover speed to bring the generator to a complete stop. ○ Turn off the field excitation. 9. Maintenance: ○ Clean and inspect the generator regularly. ○ Check windings, connections, and insulation for wear or damage. ○ Lubricate bearings as recommended by the manufacturer.
05.	Component name	Three-phase induction motor
	Quantity	01
	Ratings	Voltage: 415±10 V f=50±5% Hz n=1430 RPM P: 2.2KW Horse power: 3 H.P. Current: 4.6A P.F.: 0.83 Efficiency: 82.80%

Diagram	
Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Induction Motor: 1. Inspect the Motor: ○ Ensure the motor is securely mounted and free from visible damage. ○ Check that all terminals and connections are in proper condition. 2. Connect Power Supply: ○ Attach the three-phase power supply wires to the motor terminals. ○ Ensure the motor is properly grounded. 3. Check Starter: ○ Connect a suitable starter (e.g., DOL starter, star-delta starter) for controlled motor starting. 4. Start the Motor: ○ Switch on the power supply through the starter. ○ Gradually bring the motor up to its rated speed. 5. Monitor Operation: ○ Continuously observe voltage, current, and motor speed. ○ Check for unusual noises, vibrations, or overheating during operation. 6. Adjust Load: ○ Ensure the load connected to the motor is within its rated capacity. ○ Adjust the load as necessary for optimal performance. 7. Shut Down: ○ Turn off the power supply via the starter. ○ Allow the motor to come to a complete stop before disconnecting the load or supply. 8. Maintenance: ○ Regularly clean the motor and remove dust or debris. ○ Inspect bearings and lubricate as needed.

		○ Check winding insulation and terminal connections periodically.
06.	Component name	Single-phase Transformer
	Quantity	01
	Ratings	Rated power 10 - 2500VA Rated input voltage < 1000V Rated output voltage < 500V Frequency 50/60Hz
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Single-phase Transformer: 1. Inspect the Transformer: ○ Check for physical damage and ensure the transformer is properly mounted. ○ Verify that all terminals are clean and secure. 2. Connect the Input (Primary Side): ○ Attach the primary winding terminals to the appropriate AC power source. ○ Ensure the voltage matches the transformer's input rating. 3. Connect the Output (Secondary Side): ○ Connect the load to the secondary winding terminals as per the required voltage. ○ Confirm that the load does not exceed the rated capacity of the transformer. 4. Grounding: ○ Ensure the transformer is properly grounded for safety. 5. Power On: ○ Switch on the power supply to the primary winding. ○ Observe the output voltage to ensure it matches the expected level. 6. Monitor Operation: ○ Continuously monitor the transformer's temperature and output voltage.

		○ Check for unusual sounds, vibrations, or overheating.
		7. Shut Down: ○ Turn off the power supply to the primary winding. ○ Disconnect the load and input connections if needed.
		8. Maintenance: ○ Periodically clean the transformer to remove dust and debris. ○ Inspect the windings, terminals, and insulation for wear or damage.

TABLE 05

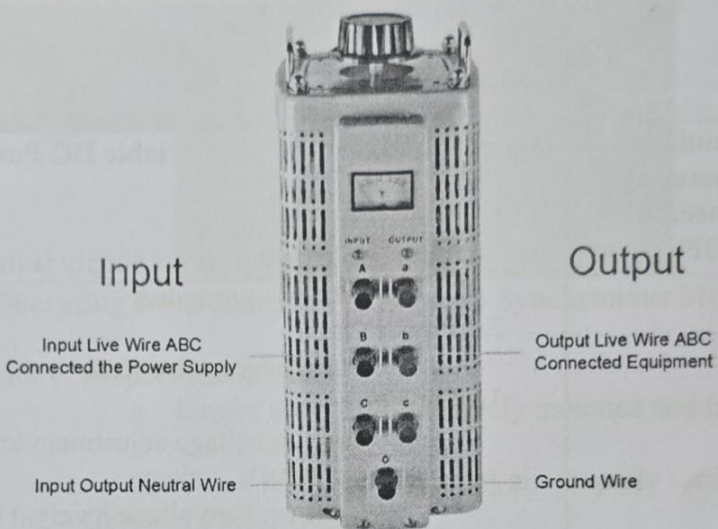
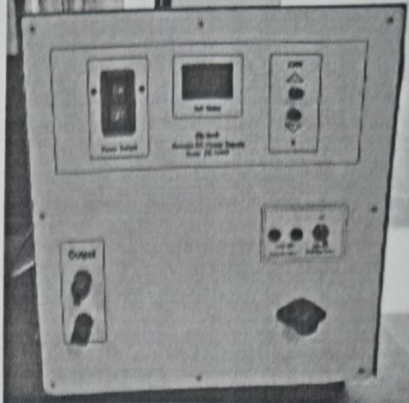
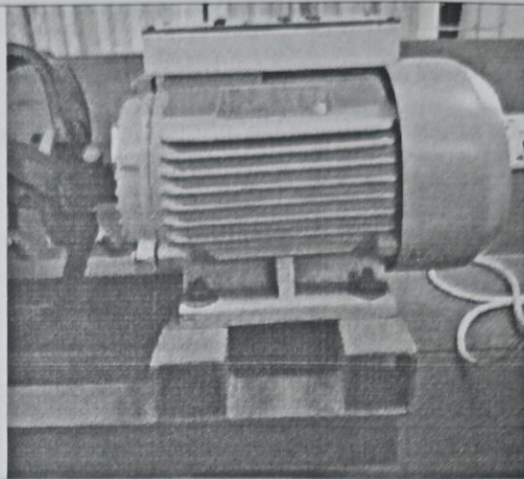
01.	Component name	Three-phase VARIAC
	Quantity	01
	Ratings	Input: 430-440V, 50HZ Output: 0-430V, 50HZ
	Connection Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase VARIAC: 1. Inspect the VARIAC for damages and ensure it is grounded. 2. Connect Input Wires: Attach live wires (A, B, C) and neutral wire (O) to the input terminals. 3. Connect Output Wires: Attach live wires (A, B, C) and ground wire to the output terminals. 4. Adjust Voltage: Rotate the control knob to set the desired output voltage. 5. Power On: Switch on the input power supply and monitor voltage using the meter. 6. Operate Equipment: Connect and operate the equipment after stabilizing the output voltage. 7. Shut Down: Turn off equipment, power off the VARIAC, and disconnect wires. 8. Maintain: Regularly clean, check for loose connections, and inspect for wear.
02.	Component name	Variable DC Power Supply
	Quantity	01
	Ratings	Input: 220V Output Variable: 0-220V

	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Variable DC Power Supply: Inspect the Device: - Ensure the power supply is in good condition and properly grounded. Power On: - Turn the power switch to the "ON" position. Set Voltage: - Use the voltage adjustment knob or buttons to set the desired DC voltage. - Confirm the voltage level on the digital voltmeter display. Connect Load: - Attach the positive and negative terminals of the output to the respective terminals of the load or equipment. Operate Equipment: - Start the connected equipment and monitor the voltage and current levels during operation. Adjust as Needed: - Fine-tune the voltage or current using the controls, if required, during operation. Shut Down: - Turn off the connected equipment. - Switch the power supply off using the power switch. - Disconnect the output terminals. Maintenance: - Periodically clean the device and check for loose connections or wear.
03.	Component name Quantity Ratings	Three-phase synchronous motor 01 P=2200W PF: 01

F=50Hz
n=1500 RPM

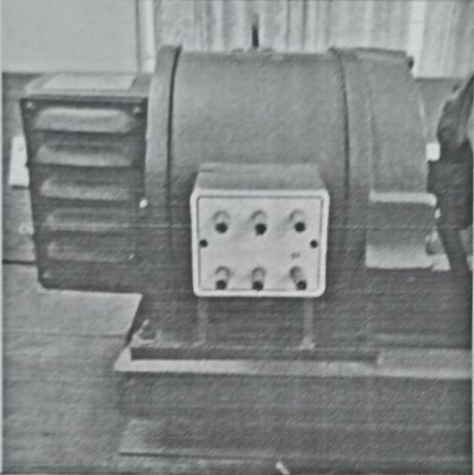
Diagram

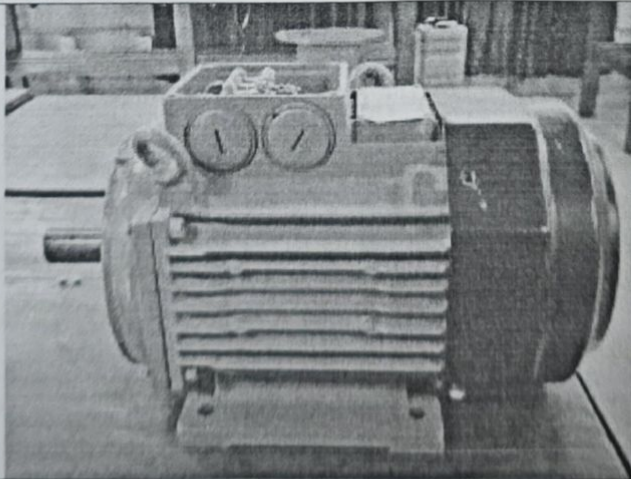


Standard
Operating
Procedure
(SOP)

Operating Procedure for Three-phase Synchronous Motor:

1. **Inspect the Motor:**
 - Ensure the motor is securely mounted and free from visible damage.
 - Check that all connections are properly secured.
2. **Connect Power Supply:**
 - Attach the **three-phase power supply wires** to the respective terminals of the motor.
 - Ensure proper grounding of the motor.
3. **Check Auxiliary Equipment:**
 - Confirm the exciter (if applicable) is operational and properly connected.
4. **Start the Motor:**
 - Apply the three-phase power supply.
 - Use a starter or control panel to gradually bring the motor to synchronous speed.
5. **Synchronize the Motor:**
 - Adjust the excitation to achieve the desired power factor and synchronize with the power supply.
6. **Monitor Operation:**
 - Continuously monitor speed, voltage, and current to ensure stable operation.
 - Listen for unusual sounds or vibrations during operation.
7. **Shut Down:**
 - Turn off the power supply and excitation system.
 - Allow the motor to come to a complete stop before disconnecting any connections.
8. **Maintenance:**
 - Regularly inspect and clean the motor.

		○ Lubricate bearings as per the manufacturer's guidelines. ○ Check winding insulation and connections periodically.
04.	Component name	Three-phase synchronous generator
	Quantity	01
	Ratings	P=1800VA PF: 01 f=50Hz n=1500 RPM
	Diagram	
	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Synchronous Generator: 1. Inspect the Generator: ○ Ensure the generator is securely mounted and free from physical damage. ○ Check that all terminals and connections are intact. 2. Connect Load: ○ Attach the load wires to the respective terminals of the generator. ○ Ensure the ground wire is properly connected. 3. Connect Prime Mover: ○ Ensure the generator is mechanically coupled to a prime mover (e.g., turbine or engine). ○ Check for proper alignment and secure coupling. 4. Start the Prime Mover: ○ Gradually increase the speed of the prime mover to bring the generator to its rated speed. 5. Excitation: ○ Apply the required field excitation to establish the magnetic field. ○ Adjust excitation to regulate voltage output.

		6. Synchronize with the Grid (if applicable): ○ Match the generator's voltage, frequency, and phase sequence with the grid before connection. ○ Close the circuit breaker to connect the generator to the grid. 7. Monitor Operation: ○ Continuously monitor voltage, current, and frequency to ensure stable operation. ○ Check for unusual sounds or vibrations. 8. Shut Down: ○ Disconnect the generator from the grid or load. ○ Gradually reduce the prime mover speed to bring the generator to a complete stop. ○ Turn off the field excitation. 9. Maintenance: ○ Clean and inspect the generator regularly. ○ Check windings, connections, and insulation for wear or damage. ○ Lubricate bearings as recommended by the manufacturer.
05.	Component name	Three-phase induction motor
	Quantity	01
	Ratings	Voltage: 415 ± 10 V $f = 50 \pm 5\%$ Hz $n = 1430$ RPM P: 2.2KW Horse power: 3 H.P. Current: 4.6A P.F.: 0.83 Efficiency: 82.80%
	Diagram	

	Standard Operating Procedure (SOP)	Operating Procedure for Three-phase Induction Motor: Inspect the Motor: - Ensure the motor is securely mounted and free from visible damage. - Check that all terminals and connections are in proper condition. Connect Power Supply: - Attach the three-phase power supply wires to the motor terminals. - Ensure the motor is properly grounded. Check Starter: - Connect a suitable starter (e.g., DOL starter, star-delta starter) for controlled motor starting. Start the Motor: - Switch on the power supply through the starter. - Gradually bring the motor up to its rated speed. Monitor Operation: - Continuously observe voltage, current, and motor speed. - Check for unusual noises, vibrations, or overheating during operation. Adjust Load: - Ensure the load connected to the motor is within its rated capacity. - Adjust the load as necessary for optimal performance. Shut Down: - Turn off the power supply via the starter. - Allow the motor to come to a complete stop before disconnecting the load or supply. Maintenance: - Regularly clean the motor and remove dust or debris. - Inspect bearings and lubricate as needed. - Check winding insulation and terminal connections periodically.
06.	Component name	Single-phase Transformer
	Quantity	01
	Ratings	Rated power 10 - 2500VA Rated input voltage < 1000V Rated output voltage < 500V Frequency 50/60Hz

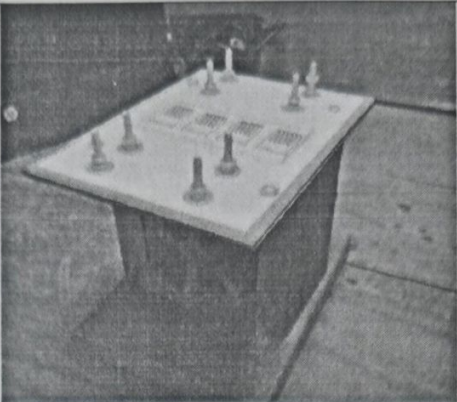
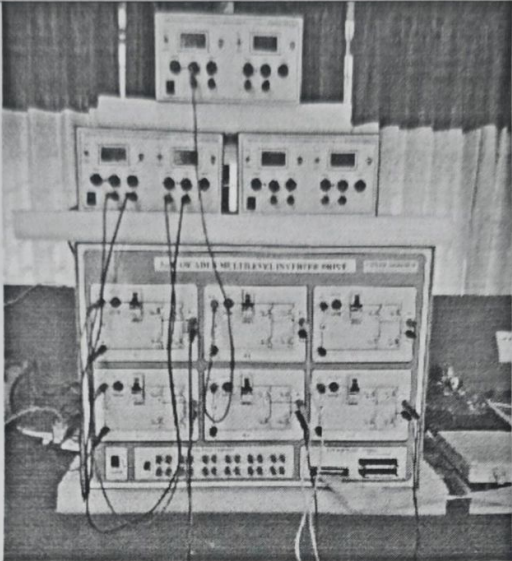
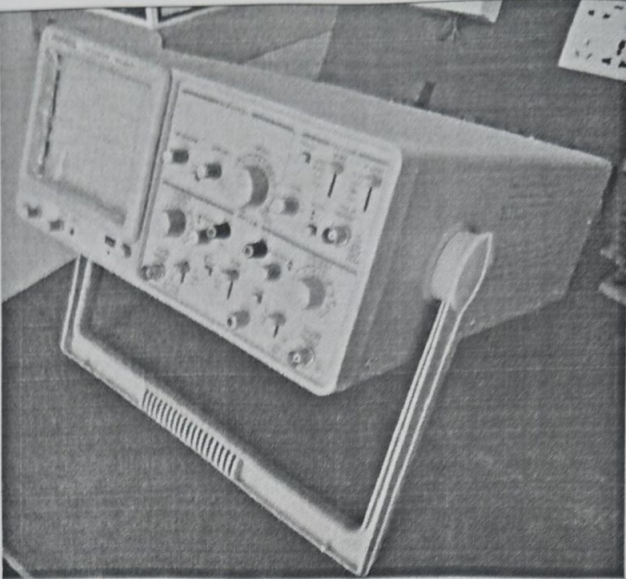
Diagram	
Standard Operating Procedure (SOP)	Operating Procedure for Single-phase Transformer: 1. Inspect the Transformer: ○ Check for physical damage and ensure the transformer is properly mounted. ○ Verify that all terminals are clean and secure. 2. Connect the Input (Primary Side): ○ Attach the primary winding terminals to the appropriate AC power source. ○ Ensure the voltage matches the transformer's input rating. 3. Connect the Output (Secondary Side): ○ Connect the load to the secondary winding terminals as per the required voltage. ○ Confirm that the load does not exceed the rated capacity of the transformer. 4. Grounding: ○ Ensure the transformer is properly grounded for safety. 5. Power On: ○ Switch on the power supply to the primary winding. ○ Observe the output voltage to ensure it matches the expected level. 6. Monitor Operation: ○ Continuously monitor the transformer's temperature and output voltage. ○ Check for unusual sounds, vibrations, or overheating. 7. Shut Down: ○ Turn off the power supply to the primary winding. ○ Disconnect the load and input connections if needed. 8. Maintenance: ○ Periodically clean the transformer to remove dust and debris. ○ Inspect the windings, terminals, and insulation for wear or damage.

TABLE 06

01.	Component name	Three-phase cascaded multilevel inverter drive
	Quantity	01
	Ratings	0-50V/0-2A (Dual)
	Diagram	
	Standard Operating Principle (SOP)	Operating Procedure for Three-phase Cascaded Multilevel Inverter Drive: 1. Inspect the System: ○ Ensure all modules and connections are securely mounted and undamaged. ○ Verify that the input and output terminals are labeled and clean. 2. Connect Power Supply: ○ Attach the three-phase power supply to the input terminals of the inverter drive. ○ Confirm that the voltage matches the rated input for the system. 3. Connect the Load: ○ Attach the motor or other load to the output terminals as per the circuit diagram. ○ Ensure the load matches the drive's rated capacity. 4. Grounding: ○ Ensure proper grounding of the inverter drive for safe operation. 5. Configure Parameters: ○ Set the desired output voltage, frequency, and modulation index using the control panel or switches.

		6. Power On: ○ Turn on the input power supply and verify the drive initializes correctly. ○ Monitor the display panel for status indicators. 7. Operate the System: ○ Gradually ramp up the output voltage using the control knobs or software interface. ○ Observe the load operation to ensure it functions as expected. 8. Monitor Operation: ○ Continuously monitor voltage, current, and temperature readings on the display. ○ Check for any unusual noises or vibrations in the load. 9. Shut Down: ○ Reduce the output voltage to zero before switching off the input supply. ○ Turn off the system and disconnect the load and power supply. 10. Maintenance: ○ Regularly clean the modules and remove dust or debris. ○ Inspect connections, switches, and components for wear or damage. ○ Verify that the control settings are calibrated properly.
02.	Component name	Oscilloscope
	Quantity	01
	Ratings	2 Channel, 50MHz Bandwidth, 7-inch WVGA (800X480) Active TFT Color Display 34 automated measurements, Dual window FFT, simultaneously monitors both the time and frequency domains, Built-in waveform limit, Dual channel frequency counter, Zoom Function, Automated, extended data logging feature, auto-ranging functions.

	Diagram	
	Standard Operating Principle (SOP)	Operating Procedure for Oscilloscope: 1. Inspect the Oscilloscope: ○ Ensure the device is clean, undamaged, and properly grounded. 2. Power On: ○ Turn on the oscilloscope using the power switch. ○ Allow the device to complete its self-initialization. 3. Connect Probes: ○ Attach the probe to the channel input (e.g., CH1 or CH2). ○ Connect the probe's ground clip to the circuit's ground. 4. Select Input Channel: ○ Choose the active input channel using the channel selection buttons. 5. Adjust Settings: ○ Set the vertical scale (volts/div) to match the signal amplitude. ○ Adjust the time base (time/div) to match the signal frequency. 6. Connect to Circuit: ○ Attach the probe tip to the point in the circuit where the signal needs to be measured. 7. Trigger Settings: ○ Adjust the trigger level to stabilize the waveform on the display. ○ Choose the appropriate trigger source and mode (e.g., edge, auto). 8. Analyze the Signal: ○ Observe the waveform on the display and make measurements as required (e.g., frequency, amplitude)

		9. Store/Record Data (if applicable): ○ Use the save or screenshot function to record the waveform for later analysis. 10. Power Off: ○ Turn off the oscilloscope after use. ○ Disconnect probes and properly store the device. 11. Maintenance: ○ Clean the screen and knobs regularly. ○ Inspect probes for damage and replace if necessary.
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